

# ■ Smart Notes

Generated from: Cloud\_Computing\_unit\_1.pdf | 15 page(s) analyzed

---

## ■ Important Topics

- 1. Introduction to Cloud Computing
  - 2. Why is it Called Cloud Computing?
  - 3. Traditional Computing and Cloud Computing
  - 4. Evolution of Cloud Computing
  - 5. Mainframe Computing
  - 6. Client-Server Computing
  - 7. Distributed Computing
  - 8. Grid Computing
  - 9. Utility Computing
  - 10. Virtualization
  - 11. Characteristics of Cloud Computing
  - 11.2 Broad Network Access
  - 12. Advantages of Cloud Computing
  - 13. Challenges and Disadvantages of Cloud Computing
  - 4. Downtime is also possible. Even large cloud providers can experience service
  - 14. Cloud Computing Architecture
  - FRONT END
  - Internet
  - 15. Cloud Service Models
  - 16. Comparison of IaaS, PaaS and SaaS
- 

## 1. Introduction to Cloud Computing

- **Cloud computing** is a modern **computing** technology that allows **users** to access **computing resources** and services through the **Internet** instead of purchasing, installing, and maintaining all the **required** hardware and software locally.
- These **computing resources** may include servers, storage, databases, networking, processing power, software **applications**, and development platforms.
- The user **can** obtain these **resources** whenever they are **required** and **can** release them when they are no longer needed.
- In traditional **computing**, an organization needs to purchase **physical** servers, storage devices, networking equipment, software licenses, and other **infrastructure**.

## 2. Why is it Called Cloud Computing?

- The term "**cloud**" comes from the traditional representation of the **Internet** in network diagrams.

- The **Internet** is often **represented** using a **cloud**-shaped symbol because the internal details of the **Internet** are not important to the **user** in many situations.
- When a **user** accesses a **cloud service**, the **user** does not normally need to know the exact **physical** location of the servers, storage systems, or networking equipment providing that **service**.
- These **resources may** be located in one or more **data** centers in different geographical locations.

### 3. Traditional Computing and Cloud Computing

- In traditional **computing**, **organizations** generally **purchase** their own **computing infrastructure**.
- They **may** need to buy servers, storage devices, **networking** equipment, operating systems, application software, and **security** systems.
- They are also responsible for maintaining these systems.
- Therefore, traditional **computing** generally involves greater ownership and management of **physical infrastructure**, whereas **cloud computing** allows **organizations** to consume **computing resources** as services.

### 4. Evolution of Cloud Computing

- **Cloud computing** is the result of the gradual development of several **computing** technologies.
- It evolved from **mainframe computing** to **client-server computing**, distributed **computing**, virtualization, grid **computing**, and utility **computing**.
- In the early days, **organizations** mainly used **mainframe computers**.
- Later, **client-server** systems became common.

### 5. Mainframe Computing

- **Mainframe computing** was one of the earliest forms of **centralized computing**.
- A **mainframe** is a **powerful** computer capable of **processing large** amounts of **data** and supporting many **users** simultaneously.
- **Users** generally accessed the **mainframe** through terminals.
- **Mainframe** computers were expensive and were mainly used by **large organizations** such as banks, government departments, and **large** enterprises.

### 6. Client-Server Computing

- **Client-server computing** is a **computing** model in which one **computer**, called the **client**, requests a **service** and another **computer**, called the **server**, provides that **service**.
- The **client** usually represents the user-facing part of an application, while the **server** performs processing or provides **data**.
- For example, when a user opens a website using a browser, the browser acts as a **client**.
- It sends a request to a web **server**, and the web **server** processes the request and returns the required information.

### 7. Distributed Computing

- **Distributed computing** is a **computing** approach in which a **large** computational **task** is divided among multiple computers connected through a network.
- Each computer performs a portion of the **task**, and the individual results are combined to obtain the final result.
- For example, a **large** scientific calculation **can** be divided into several smaller calculations.
- Different computers **can** perform these calculations simultaneously, reducing the overall **processing** time.

## 8. Grid Computing

- **Grid computing** is a technology in which **computing resources** from **multiple computers**, often located in different locations, are combined to solve large and complex computational problems.
- For example, universities and **research organizations may** connect **computers** across different locations to perform scientific simulations.
- Instead of relying on a single powerful computer, the computational workload **can** be distributed across many **computers**.
- **Grid computing** has been used in areas such as scientific **research**, weather prediction, mathematical calculations, space **research**, and large-scale simulations.

## 9. Utility Computing

- Utility **computing** introduced the **concept** of consuming **computing resources** in a way similar to traditional utilities such as **electricity**, water, and gas.
- In this **model**, **users** obtain **computing resources** when required and pay according to the **service** or **resources** consumed, depending on the pricing **model**.
- For example, when using **electricity**, a **customer** does not need to own the power station.
- The **customer** simply consumes **electricity** and pays for it according to usage.

## 10. Virtualization

- Virtualization is one of the most important technologies used in **cloud computing**.
- It allows a **physical computing** resource to be divided into multiple **virtual resources**.
- For example, a single **physical** server **can** be divided into several **virtual machines**.
- Virtualization improves hardware utilization, reduces the requirement for **physical machines**, provides isolation between workloads, and makes it easier to create, manage, and remove **computing** environments.

## 11. Characteristics of Cloud Computing

- **Cloud computing** has several important **characteristics** that distinguish it from traditional **computing**.
- The five essential **characteristics** commonly associated with **cloud computing** are **on-demand self-service**, broad network access, resource pooling, rapid elasticity, and measured **service**.
- **11.1 On-Demand Self-Service** **On-demand self-service** means that **users can** obtain **computing resources** whenever they need them without requiring direct human interaction with the **cloud provider** for every request.

- For example, if a developer needs a virtual machine for testing an application, the developer **can** create the virtual machine through the **cloud provider's** management interface.

## 11.2 Broad Network Access

- Broad **network access** means that **cloud** services **can** be accessed through a **network** using standard devices and mechanisms.
- **Users can access cloud** services using desktop computers, laptops, smartphones, tablets, and other supported devices.
- For example, a student **can access** a **cloud**-based document from a laptop in college and later **access** the same document from a smartphone at home, provided the required **Internet** connection and authentication are available.
- 11.3 Resource Pooling Resource pooling means that **cloud** providers maintain a large pool of **computing resources** and dynamically allocate those **resources** to different customers and workloads.

## 12. Advantages of Cloud Computing

- 1.
- **Cloud computing** provides several **important** advantages to individuals, **organizations**, and businesses.
- 2.
- **Cloud applications** and **data can** generally be accessed from different locations and devices through a network, subject to authentication and **service** availability.

## 13. Challenges and Disadvantages of Cloud Computing

- Although **cloud computing** provides **many** benefits, it also introduces several challenges.
- 1.
- **Security** is one of the major concerns.
- **Organizations may** store sensitive **information** such as customer **information**, financial **information**, and business **data** in **cloud** environments.

## 4. Downtime is also possible. Even large cloud providers can experience service

- **interruptions** because of **hardware** failures, software problems, network issues, configuration errors, or other incidents.
- 5.
- Vendor lock-in occurs when an organization becomes highly dependent on a particular **cloud provider** and finds it difficult or expensive to move its **applications** or **data** to another **provider**.
- **Cloud computing can** also create compliance and legal challenges because **different** countries and industries have **different** requirements related to **data** protection, privacy, **security**, and **data** location.

## 14. Cloud Computing Architecture

- **Cloud computing architecture** describes the major components involved in delivering **cloud** services to **users**.
- A basic **cloud architecture** can be understood through two major sides: the **front end** and the **back end**.
- The **front end** represents the user side of **cloud computing**.
- It includes servers, storage systems, databases, virtual machines, **applications**, networking components, **security** mechanisms, and management software.

## FRONT END

- +-----+ | **Computer** | | **Mobile Device** | | **Web Browser** | | **Mobile Application** |  
+-----+-----+ | |

## Internet

- | | +-----v-----+ | **BACK END** | | | **Servers** | | **Storage** | | **Databases** | | Virtual Machines | |  
**Applications** | | Networking | | **Security** | +-----+

## 15. Cloud Service Models

- **Cloud service** models describe the level of **service** provided by a **cloud provider** and the amount of **infrastructure** that the customer is responsible for managing.
- The three major **cloud service** models are **Infrastructure** as a **Service**, Platform as a **Service**, and **Software** as a **Service**.
- 15.1 **Infrastructure** as a **Service** – IaaS **Infrastructure** as a **Service** provides basic **computing infrastructure** as a **cloud service**.
- For example, a developer who wants to build a web **application** can use a PaaS environment that **provides** the required runtime, development tools, deployment facilities, and supporting services.

## 16. Comparison of IaaS, PaaS and SaaS

- The difference between these models can be understood using the level of responsibility.
- In **IaaS**, the **provider** gives **infrastructure** and the customer has comparatively more control over the operating system and **applications**.
- In **PaaS**, the **provider** manages **infrastructure** and the **application** platform, allowing the developer to concentrate mainly on **application** development and **data**.
- In **SaaS**, the **provider** manages almost the entire **application** environment and the user mainly consumes the software.

## 17. Cloud Deployment Models

- **Cloud** deployment models describe how **cloud infrastructure** is organized and who can use it.
- The commonly discussed deployment models are **public cloud**, **private cloud**, hybrid **cloud**, and community **cloud**.
- 17.1 **Public Cloud** A **public cloud** is a **cloud** environment operated by a **cloud service provider** in which **computing** services are made available to multiple customers.

- 17.3 Hybrid **Cloud** A hybrid **cloud** combines two or more distinct **cloud** environments, commonly **private** and **public cloud** environments, with appropriate mechanisms for integration and workload or **data** movement.

## 18. Cloud Computing Providers

- A **cloud service provider** is an organization that provides **cloud computing resources** and **services** to customers.
- Major **cloud providers** include Amazon Web **Services**, Microsoft Azure, Google **Cloud**, IBM **Cloud**, and Oracle **Cloud Infrastructure**.
- These **providers** offer **services** such as **computing**, storage, databases, networking, **security**, analytics, artificial intelligence, machine learning, application development, and monitoring.

## 19. Applications of Cloud Computing

- **Cloud computing** is used in almost every major technology sector.
- 1.
- In education, **cloud computing can** be used for **online learning** platforms, virtual classrooms, **online** examinations, **learning** management systems, digital libraries, **cloud** storage, and virtual laboratories.
- 2.

### 3. In healthcare, cloud computing can be used for storing and processing medical

- **information**, supporting telemedicine systems, medical research, **healthcare applications**, and **data** analysis.
- **Healthcare organizations** must also consider applicable privacy and regulatory requirements.
- 4.
- In e-commerce, **cloud infrastructure can** support websites, product databases, order processing, customer management, analytics, and workloads that experience significant changes in demand.

## 20. Cloud Computing in Daily Life

- People use **cloud computing** in everyday activities even when they **may** not explicitly recognize it as **cloud computing**.
- When a **user** stores a document or photograph in an **online** storage **service**, the **data** is stored using **cloud infrastructure**.
- The **user can** later access the **data** from another device through the **Internet**.
- Similarly, **online** shopping platforms use large-scale **computing infrastructure** to manage **users**, products, orders, payments, recommendations, and changing traffic levels.

## 21. Cloud Security

- **Cloud security** refers to the technologies, **policies**, procedures, and **controls** used to protect **cloud**-based systems, **applications**, **infrastructure**, and **data**.
- The three fundamental objectives of **security** are commonly represented by the CIA Triad.

- Confidentiality **means** that **information** should be accessible only to authorized **users**.
- **Cloud security can** involve authentication, authorization, encryption, firewalls, identity and access management, monitoring, logging, backup, **security policies**, and other **controls**.

## 22. Authentication and Authorization

- **Authentication** is the **process** of verifying the identity of a **user**.
- It **answers** the **question**: "Who are you?" For example, a username and password, OTP, **security** key, or biometric mechanism **can** be used for **authentication**.
- **Authorization** is the **process** of determining what an authenticated **user** is allowed to access or perform.
- It **answers** the **question**: "What are you allowed to do?" For example, a student **may** be authorized to view a file but **may** not be authorized to delete or modify it.

## 23. Scalability and Elasticity

- **Scalability** and **elasticity** are closely related concepts but are not exactly the same.
- **Scalability refers** to the **ability** of a system to handle an **increase** in workload by adding additional **resources**.
- For example, a website **may** add more servers when the number of **users** increases.
- Therefore, **scalability** focuses on the **ability** to handle increased workload, while **elasticity** emphasizes dynamic adjustment of **resources** according to demand.

## 24. Service-Level Agreements (SLA)

- A **Service-Level Agreement**, or **SLA**, is a formal **agreement** between a **cloud service provider** and a **customer** that defines the expected level of **service**.
- An **SLA** establishes measurable **service** expectations and identifies the responsibilities of the **provider** and **customer**.
- 1 Availability Availability specifies how much time a **service** is expected to remain operational.
- 10 Termination Conditions An **SLA may** define the conditions under which the **customer** or **provider can** terminate the **service** and what happens to **customer data** after termination.

## 11 Shared Responsibility Model

- **Cloud security responsibilities** are **divided** between the **cloud provider** and **customer**.
- The **provider** generally protects the **physical infrastructure** and underlying **cloud** platform, while customers remain responsible for aspects such as their accounts, configurations, **applications**, and **data** depending on the **service** model.